Genotype likelihoods, allele frequencies, and genotyping from NGS data

Tyler Linderoth
Physalia lcWGS course 2026

Tentative plan

Day 1 Recap. Fastq to BAM (30 min)
5 min break
Lecture 1 + GL exercise intro. Motivate probabilistic approach and genotype likelihoods. (45 min)
15 min break
Exercise 1. ANGSD I/O, common options, and genotype likelihoods. (45 min)
5 min break
Lecture 2 + AF exercise intro. Maximum likelihood allele frequency estimation. (35 min)
30 min break
Exercise 2. Estimate allele frequencies and call SNPs with ANGSD. (45 min)
5 min break
Lecture 3 + genotyping exercise intro. Genotype probabilities and calling. (40 min)
15 min break
Exercise 4. Calculate genotype posterior probabilities and call genotypes with ANGSD. (45 min)

Goals

- Understand why naively calling genotypes with low coverage data is not a good idea.

Gain sufficient understanding of the theory behind the probabilistic approach to genotyping and estimation of related quantities to be able to think critically about how different data processing practices could affect results.

Know how to read in data and apply basic QC criteria in ANGSD to:

- Calculate genotype likelihoods.
- Estimate allele frequencies.
- Call SNPs.
- Calculate genotype posterior probabilities.
- Call genotypes (if necessary).

Understand the contents of the ANGSD output files.

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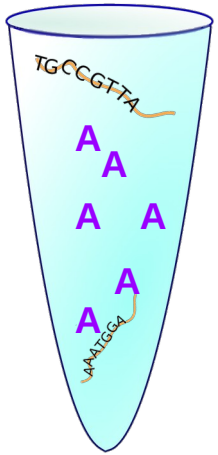
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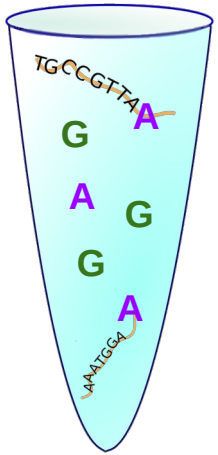
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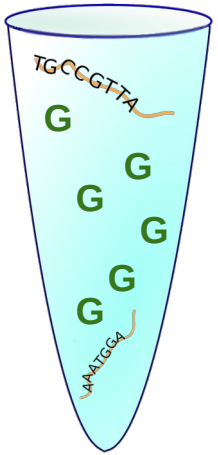
Why take a probabilistic approach?



The library for an individual homozygous for the **A** allele will consist only of **A**s.

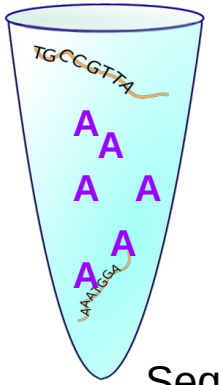


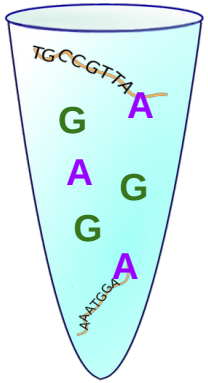
The library for a heterozygous individual at a site contains both **A**s and **G**s.

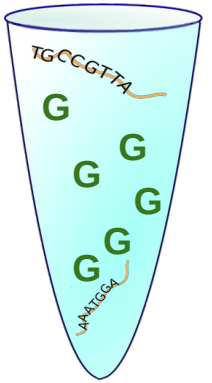


The library for an individual homozygous for the **G** allele will consist only of **G**s.

Why take a probabilistic approach?







Sequence to average depth of 4x.



$$\text{Depth} \sim \text{Poisson}(\lambda = 4)$$

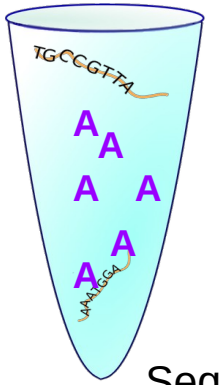
$$\text{E}[\text{depth}] = \lambda$$

$$\text{Var}[\text{depth}] = \lambda$$



Expect 4 reads, all with **A**s at this ref position.

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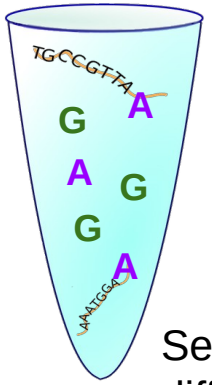
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Sequencing (sampling) the two different alleles is just like flipping a coin.

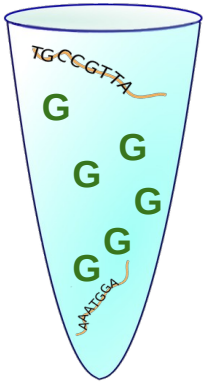
$$\#\,\text{A alleles} \sim \text{Binomial}(n\,reads, p = 0.5)$$

$$\text{E}[\text{A depth}] = np = 0.5n$$

$$\text{Var}[\text{A depth}] = np(1-p) = 0.25n$$

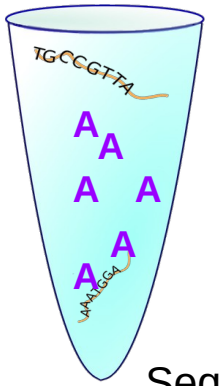


Expect 2 A alleles and 2 G alleles





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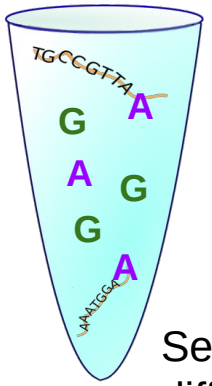
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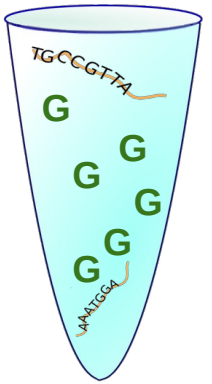
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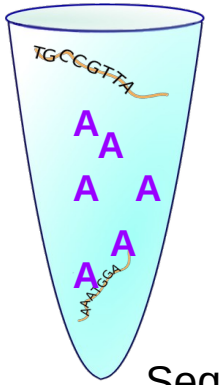


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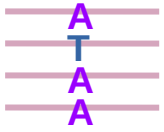


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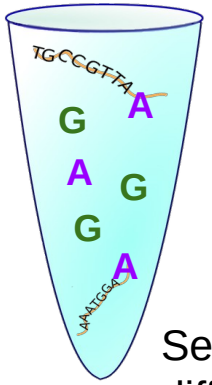
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Sequencing error. Rates of ~0.1% for older platforms.





Sequencing (sampling) the two different alleles is just like flipping a coin.

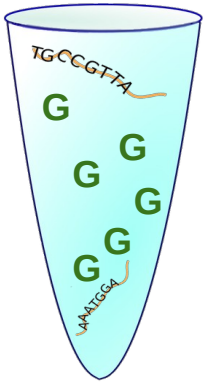
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Expect 2 A alleles and 2 G alleles





SAMtools mpileup representation of sequencing data for two individuals



	Individual 1 data		Individual 2 data		
6	.....,	DEGEGG	9	,.,,.,,.,	DABGIIIII
6	.....,	DEGEGG	9	,.,,.,,.,	DABGIIIII
6	.....,	DEGEGG	9	,.,,.,,.,	DABGIIIII
6	.....,	DEGEGG	10	,.,,.,,.,^].	DABGIIIIIE
6	.....,	DEGEGG	10	,.,,.,,.,.	DABGIIIIII
6	.....,	DEGEGG	10	,.,,.,,.,.	DABGIIIIII
6	.....,	DEGEGG	10	,.,,.,,.,.	DABGIIIIII
7	.....,^].	DEGEGGE	10	,.,,.,,.,.	DABGIIIIII
7	.....,.	DEGEGGG	10	,.,,.,,.,.	DABGIIIIII
8	.....,.^],	DEGEGGGB	10	,.,,.,,.,.	DABGIIIIII
8	.....,.,	DEGEGGGB	10	,.,,.,,.,.	DABGIIIIII
8	.....,.,	DEGEGGGB	10	,.,,.,,.,.	DABGIIIIII
8	.....,.,	DEGEGGGB	10	,.,,.,,.,.	DABGIIIIII
9	.....,.,^],	DEGEGGGBE	10	,.,,.,,.,.	DABGIIIIII
9	.G...gG,,	DEGEGGGBG	10	,.,,.,,.,.	D3BGIIIIII
9	.....,.,,	DEGEGGGBG	10	,.,,.,,.,.	D3BGIIIIII

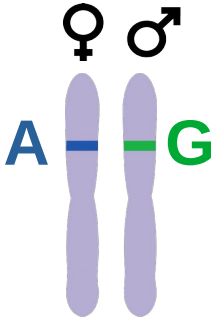
Each row is a different site in the reference genome

reads

base IDs

base qualities

Example sequencing data for one individual at Chr1:472

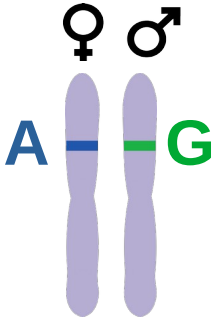


Maternally and paternally inherited chromosome 1 of a diploid individual.



Scaffold	position	read bases	base qualities
Chr1	472	AAG	DEG

Example sequencing data for one individual at Chr1:472
Sequencing error encoding.



Maternally and paternally inherited chromosome 1 of a diploid individual.



Scaffold	position	read bases	base qualities
Chr1	472	AAG	DEG

↓

ASCII character “D” = decimal value 68

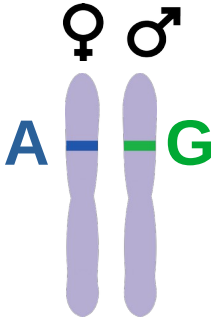
↓

68 – 33 = base quality of 35

↓

$$\text{P(error)} = \epsilon = 10^{\frac{-35}{10}} = 0.00032$$

Example sequencing data for one individual at Chr1:472





Scaffold	position	read bases	base qualities
Chr1	472	AAG	DEG

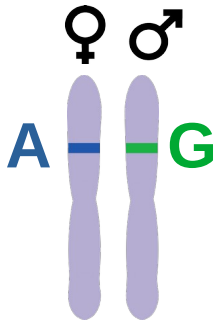
Maternally and paternally inherited chromosome 1 of a diploid individual.

This individual could have any of the following 10 genotypes (we can only see the sequencing data):

AA, AC, AG, AT, CG, CC, CT, GG, GT, TT

How do we figure out which genotype they are most likely to have based on the observed sequence data?

Genotype likelihoods



true genotype



Scaffold	position	read bases	base qualities
Chr1	472	AAG	DEG

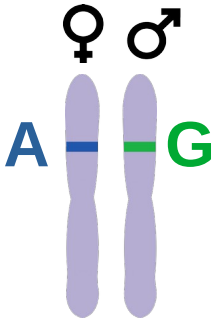
$$\mathrm{P}\left(\mathrm{Data}\middle|\mathrm{Genotype}=bh\right) = L\left(\mathrm{Genotype}\,bh\right) = \text{likelihood of genotype } bh$$

$$b,h \in \{\mathrm{A},\mathrm{C},\mathrm{G},\mathrm{T}\}$$

Possible genotypes: AA, AC, AG, AT, CG, CC, CT, GG, GT, TT

What is the likelihood that the individual's genotype is AC?

Possible genotypes: AA, **AC**, AG, AT, CG, CC, CT, GG, GT, TT



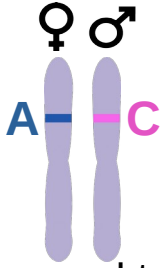
true genotype



Scaffold	position	read bases	base qualities
Chr1	472	AAG	DEG

P(observed *i*-th read | A allele) P(observed *i*-th read | C allele)

$$P(\text{Data}|\text{Genotype}=\text{AC})=\square\times\square+\square\times\square$$

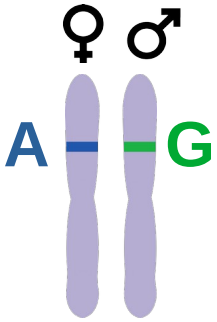


Assumed true genotype

Probability of sampling maternal chromosome (A allele)

Probability of sampling paternal chromosome (C allele)

What is the likelihood that the individual's genotype is AC?



true genotype

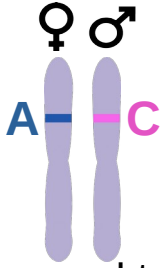


Scaffold	position	read bases	base qualities
Chr1	472	**A**AG	DEG

$$P(\text{Data}|\text{Genotype}=AC)=$$ ☐ × ☐ + ☐ × ☐ × ☐

P(observed read A | A allele)

P(observed read A | C allele)

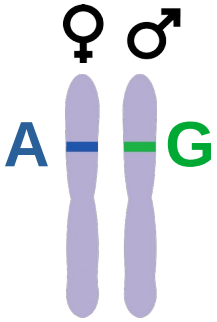


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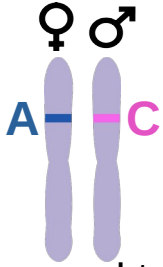


Scaffold	position	read bases	base qualities
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$$P(\text{Data}|\text{Genotype}=AC)=\frac{1}{2}\times\square+\square\times\square\times\square$$

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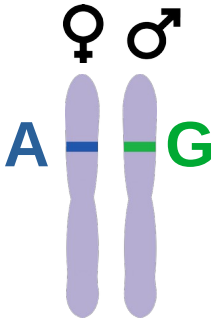


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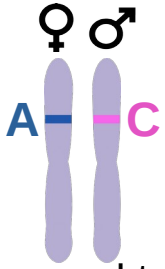
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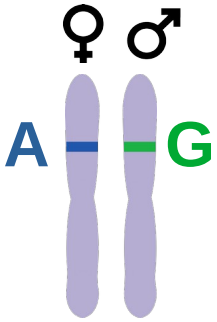
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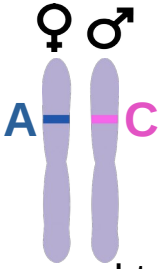
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$$P(\text{Data}|\text{Genotype}=\text{AC}) = \frac{1}{2} \times (1-\epsilon_1) + \square \times \square \times \square$$

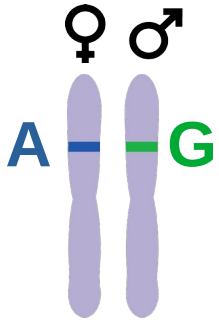
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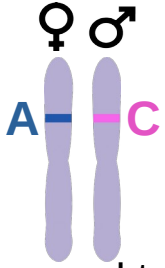
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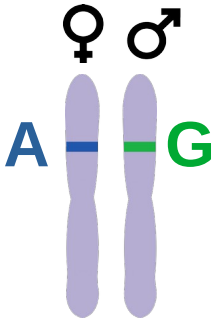
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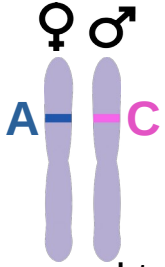
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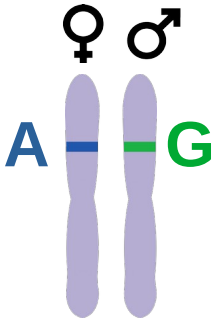
P(observed read A | C allele)

$$P(\text{Data}|\text{Genotype}=\text{AC}) = \frac{1}{2} \times (1-\epsilon_1) + \frac{1}{2} \times \square \times \square$$

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Scaffold	position	read bases	base qualities
Chr1	472	AAG	DEG

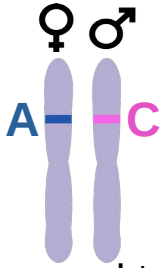
P(observed read A | A allele)

P(observed read A | C allele)

$$P(\text{Data}|\text{Genotype}=\text{AC}) = \frac{1}{2} \times (1-\epsilon_1) + \frac{1}{2} \times \square \times \square$$

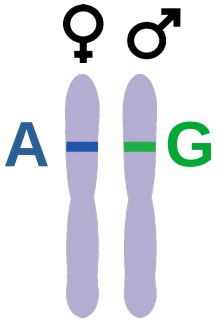
Probability of sampling maternal chromosome (A allele)

Probability of sampling paternal chromosome (C allele)



Assumed true genotype

What is the likelihood that the individual's genotype is AC?



true genotype



Scaffold	position	read bases	base qualities
Chr1	472	AAG	DEG

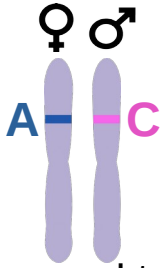
P(observed read A | A allele)

P(observed read A | C allele)

$$P(\text{Data}|\text{Genotype}=\text{AC}) = \frac{1}{2} \times (1-\epsilon_1) + \frac{1}{2} \times \epsilon_1 \times \frac{1}{3}$$

Probability of sampling maternal chromosome (A allele)

Probability of sampling paternal chromosome (C allele)



Assumed true genotype

What is the likelihood that the individuals genotype is AC?

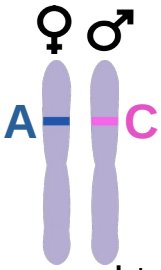


Assuming equal probability of changing to any of the possible 3 erroneous bases.

P(observed read A | A allele)

P(observed read A | C allele)

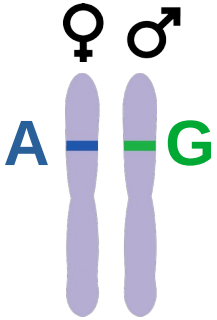
$$P(\text{Data}|\text{Genotype}=\text{AC}) = \boxed{\frac{1}{2}} \times \boxed{1-\epsilon_1} + \boxed{\frac{1}{2}} \times \boxed{\epsilon_1} \times \boxed{\frac{1}{3}}$$



Assumed true genotype

Probability of sampling maternal chromosome (A allele)

Probability of sampling paternal chromosome (C allele)

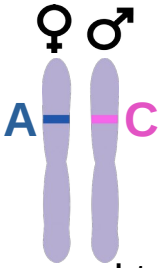


true genotype



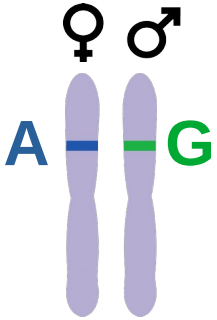
Scaffold	position	read bases	base qualities
Chr1	472	AAG	DEG

$$P(\text{Data}|\text{Genotype}=\text{AC}) = \left[\frac{1}{2}\times(1-\epsilon_1) + \frac{1}{2}\times\epsilon_1\times\frac{1}{3}\right]\times\left[\frac{1}{2}\times(1-\epsilon_2) + \frac{1}{2}\times\epsilon_2\times\frac{1}{3}\right]$$



Assumed true genotype

When multiple independent events occur, we multiply their probabilities. In this case, the sequencer sampled an A allele *and* it sampled another A allele.



true genotype



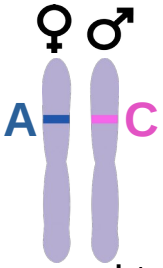
Scaffold	position	read bases	base qualities
Chr1	472	AAG	DEG

$$P(\text{Data}|\text{Genotype}=\text{AC})=$$

$$\left[\frac{1}{2}\times(1-\epsilon_1)+\frac{1}{2}\times\epsilon_1\times\frac{1}{3}\right]\times$$

$$\left[\frac{1}{2}\times(1-\epsilon_2)+\frac{1}{2}\times\epsilon_2\times\frac{1}{3}\right]\times$$

$$\left[\frac{1}{2}\times\epsilon_3\times\frac{1}{3}+\frac{1}{2}\times\epsilon_3\times\frac{1}{3}\right]$$



Assumed true genotype

General genotype likelihood expression

$$\mathrm{P}\left(\mathrm{Data} \mid \mathrm{Genotype} = bh\right) = \prod_{i=1}^{r\ \mathrm{reads}} \left(\frac{L_b^{(i)}}{2} + \frac{L_h^{(i)}}{2} \right)$$

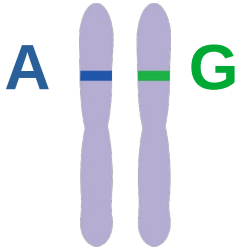
$$b, h \in \{\mathrm{A}, \mathrm{C}, \mathrm{G}, \mathrm{T}\}$$

$$L_b^{(i)} = \mathrm{P}\left(\mathrm{observed\ read} = x_i \mid \mathrm{assumed\ true\ allele} = b\right)$$

$$L_h^{(i)} = \mathrm{P}\left(\mathrm{observed\ read} = x_i \mid \mathrm{assumed\ true\ allele} = h\right)$$

$$\begin{cases} \frac{\epsilon_i}{3} & \text{if } b, h \neq x_i \\ 1 - \epsilon_i & \text{if } b, h = x_i \end{cases}$$

Lets assume the same base quality for each observed allele for simplicity of the example.





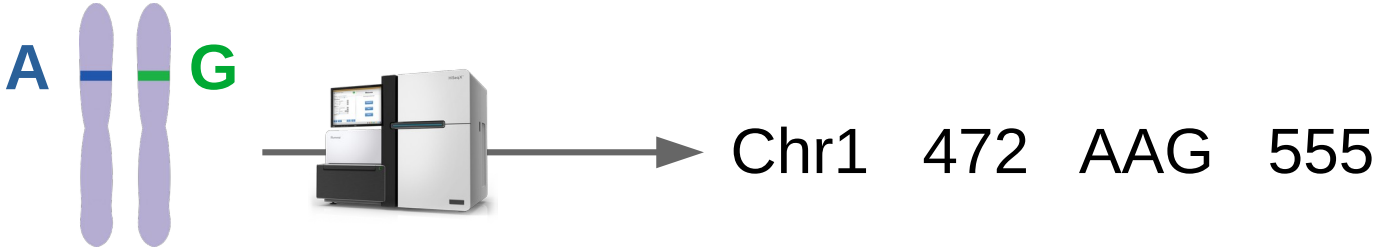
Chr1 472 AAG 555

ASCII character “5” = decimal value 53

53 – 33 = base quality of 20

$$\epsilon = 10^{\frac{-20}{10}} = 0.01$$

Representation of genotype likelihoods in ANGSD

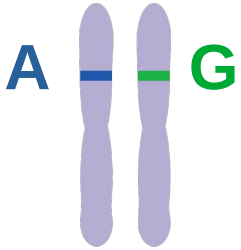


Instead of a single genotype, AG, we have a distribution over all possible genotypes:

Genotype	AA	AC	AG	AT	CC	CG	CT	GG	GT	TT
Log_{10} likelihood	-2.49	-3.08	-0.91	-3.08	-7.43	-5.26	-7.43	-4.96	-5.26	-7.43

$$\epsilon = 10^{\frac{-20}{10}} = 0.01$$

Representation of genotype likelihoods in ANGSD





Chr1 472 AAG 555

Instead of a single genotype, AG, we have a distribution over all possible genotypes:

Genotype	AA	AC	**AG**	AT	CC	CG	CT	GG	GT	TT
Log_{10} likelihood	-2.49	-3.08	**-0.91**	-3.08	-7.43	-5.26	-7.43	-4.96	-5.26	-7.43

Maximum likelihood estimate of the genotype

$$\epsilon = 10^{\frac{-20}{10}} = 0.01$$

Exercise. Calculate genotype likelihoods with ANGSD.

Estimating allele frequencies

When genotypes are known, allele frequencies can be calculated by simply counting alleles.

	Ind1	Ind2	Ind3	Ind4	Ind5	Ind6	Ind7	Ind8	Ind9	Ind10		Minor allele frequency (MAF)
Site 1	0	0	1	0	0	0	1	1	0	0	=	0.15
Site 2	0	2	0	0	1	0	0	0	1	0	=	0.20

Genotype notation

0 = zero minor alleles
1 = one minor alleles
2 = two minor alleles

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Site 1	0	0	1	0	0	0	1	1	0	0	=	0.15
Site 2	0	2	0	0	1	0	0	0	1	0	=	0.20

But how do you estimate allele frequencies when you have a distribution of genotype likelihoods?

Maximum likelihood estimation of allele frequencies

$$\mathrm{P}(\mathrm{Data}|f) = \prod_{i=1}^{n\ \mathrm{individuals}} \sum_{g \in \{0,1,2\}} \mathrm{P}(D_i|\mathrm{Genotype}_i = g)\,\mathrm{P}(\mathrm{Genotype}_i = g|f)$$

$D_i =$ sequencing data for individual i

$f =$ population minor allele frequency

Genotype notation

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The likelihood of genotype *g* for individual *i* calculated as shown previously.

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Genotype notation

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The likelihood of genotype *g* for individual *i* calculated as shown previously.

Hardy-Weinberg frequency of genotypes.

$D_i =$ sequencing data for individual i

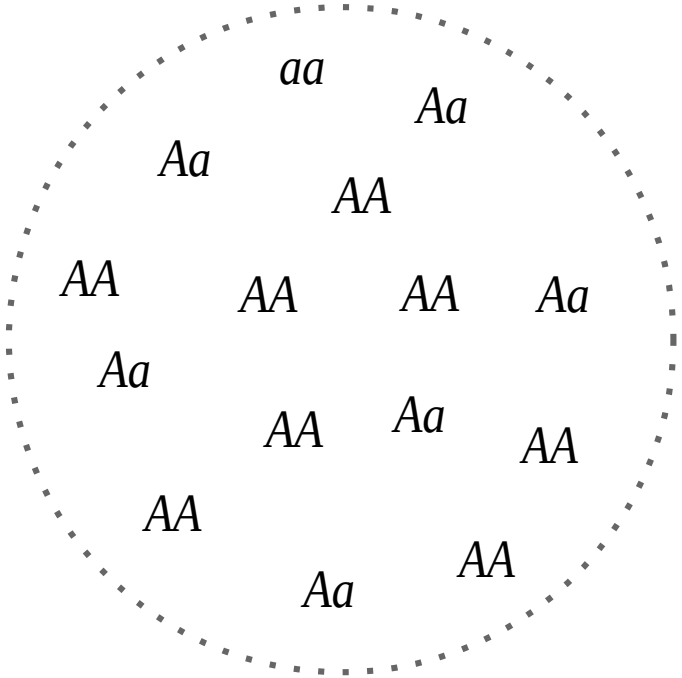
$f =$ population minor allele frequency

Genotype notation

0 = zero minor alleles
1 = one minor alleles
2 = two minor alleles

Genotype frequencies under Hardy-Weinberg Equilibrium (HWE)

$$\mathrm{P}(\mathrm{Data}|f) = \prod_{i=1}^{n\ \text{individuals}} \sum_{g \in \{0,1,2\}} \mathrm{P}(D_i|\mathrm{Genotype}_i = g)\,\mathrm{P}(\mathrm{Genotype}_i = g|f)$$

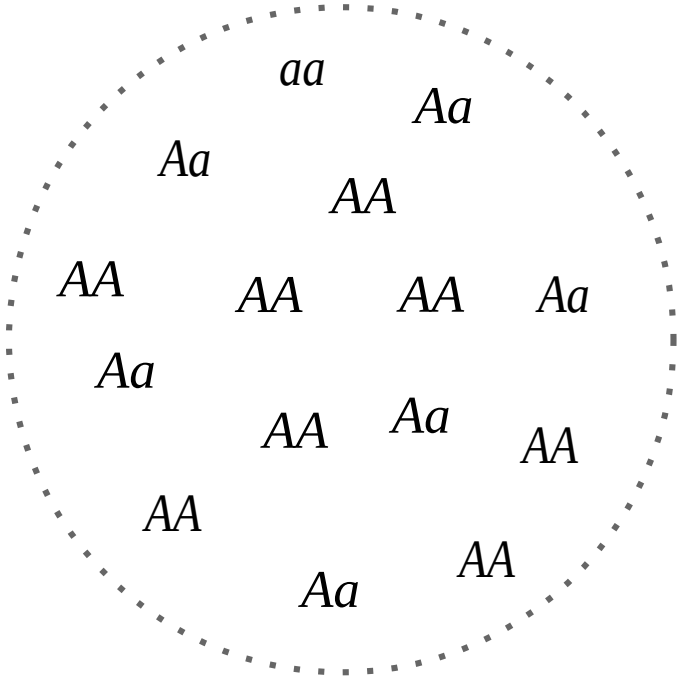


An “infinitely” large population of sexually reproducing diploid organisms segregating for alleles *A* and *a*, and for which

- Mating is random.
- Generations are nonoverlapping.
- Allele frequencies are the same in males and females.
- No migration, mutation, or selection.

Genotype frequencies under Hardy-Weinberg Equilibrium (HWE)

$$\mathrm{P}(\mathrm{Data}|f) = \prod_{i=1}^{n\ \text{individuals}} \sum_{g\in\{0,1,2\}} \mathrm{P}(D_i|\mathrm{Genotype}_i = g)\,\mathrm{P}(\mathrm{Genotype}_i = g|f)$$



Offspring are formed through independent draws of gametes from this population, expressed as $(f_A + f_a)^2$, which expanded yields:

$$f_{AA} = f_A^2 = (1-f_a)^2 = \mathrm{P}(\mathrm{Genotype} = AA|f_a)$$

$$f_{Aa} = 2f_A f_a = 2(1-f_a)f_a = \mathrm{P}(\mathrm{Genotype} = 1|f_a)$$

$$f_{aa} = f_a^2 = \mathrm{P}(\mathrm{Genotype} = 2|f_a)$$

Maximum likelihood estimation of allele frequencies

$$\mathrm{P}(\mathrm{Data}|f) = \prod_{i=1}^{n\ \text{individuals}} \sum_{g\in\{0,1,2\}} \mathrm{P}(D_i|\mathrm{Genotype}_i = g)\,\mathrm{P}(\mathrm{Genotype}_i = g|f)$$

When one outcome occurs *or* a different outcome occurs we sum their probabilities. In this case the genotype can be 0 *or* 1 *or* 2.

Marginal probabilities are used to account for uncertainty in various quantities associated with low coverage sequencing. In general, for random variables *X* and *Y*

$$\mathrm{P}(X = x) = \sum_{y} \mathrm{P}(X = x|Y = y)\,\mathrm{P}(Y = y)$$

Maximum likelihood estimation of allele frequencies

$$\mathrm{P}(\mathrm{Data}|f) = \prod_{i=1}^{n\ \mathrm{individuals}} \sum_{g\in\{0,1,2\}} \mathrm{P}(D_i|\mathrm{Genotype}_i = g)\,\mathrm{P}(\mathrm{Genotype}_i = g|f)$$

Summing over the possible genotypes accounts for genotyping uncertainty.

Marginal probabilities are used to account for uncertainty in various quantities associated with low coverage sequencing. In general, for random variables *X* and *Y*

$$\mathrm{P}(X = x) = \sum_{y} \mathrm{P}(X = x|Y = y)\,\mathrm{P}(Y = y)$$

Maximum likelihood estimation of allele frequencies

$$\mathrm{P}(\mathrm{Data}|f) = \prod_{i=1}^{n\ \text{individuals}} \sum_{g\in\{0,1,2\}} \mathrm{P}(D_i|\mathrm{Genotype}_i = g)\,\mathrm{P}(\mathrm{Genotype}_i = g|f)$$

The value of *f* that maximizes the likelihood function above yields a **maximum likelihood estimate of *f*** :

$$\hat{f} = \mathrm{argmax}_f\, \mathrm{P}(Data|f)$$

Using the ML allele frequency estimate to identify polymorphic sites

$$\mathrm{P}(\mathrm{Data}|f) = \prod_{i=1}^{n\ \text{individuals}} \sum_{g \in \{0,1,2\}} \mathrm{P}(D_i|\mathrm{Genotype}_i = g)\,\mathrm{P}(\mathrm{Genotype}_i = g|f)$$

$$\hat{f} = \mathrm{argmax}_f\, \mathrm{P}(Data|f)$$ ← **maximum likelihood estimate of *f***

Probability of the sequencing data when $f = 0$, i.e., the site is monomorphic (null case).

$$\lambda = -2\ln\left(\frac{\mathrm{P}(\mathrm{Data}|f_0)}{\mathrm{P}(\mathrm{Data}|\hat{f})}\right) = -2[\ln(\mathrm{P}(\mathrm{Data}|f_0)) - \ln(\mathrm{P}(\mathrm{Data}|\hat{f}))]$$

$$\lambda \sim \chi^2(1\ \text{degree of freedom})$$ → Call SNPs at a given level of statistical confidence.

Exercise. Estimate allele frequencies with ANGSD.

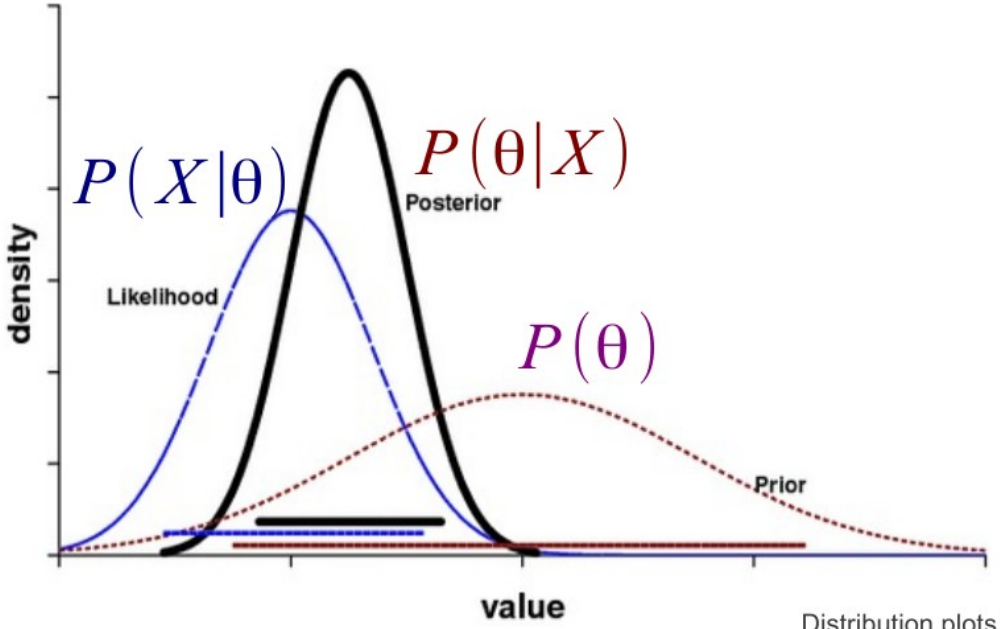
Can we use other information from our data to further increase genotyping accuracy?

6	.....,	DEGEGG	9	,.,,.,,.,	DABGIIIII	5	,,.,.	>AB/A	6	.,,...	FGG
6	.....,	DEGEGG	9	,.,,.,,.,	DABGIIIII	5	,,.,.	>ABDA	6	.,,...	3GGBGB
6	.....,	DEGEGG	9	,.,,.,,.,	DABGIIIII	5	,,.,.	>ABDA	6	.,,...	3GGBGB
6	.....,	DEGEGG	10	,.,,.,,.,^].	DABGIIIIIE	5	,,.,.	>AB/A	6	.,,...	3GGBGB
6	.....,	DEGEGG	10	,.,,.,,.,.	DABGIIIIII	5	,,.,.	>AB/A	6	.,,...	3GGBGB
6	.....,	DEGEGG	10	,.,,.,,.,.	DABGIIIIII	5	,,.,.	>ABDA	6	.,,...	BGGBGB
6	.....,	DEGEGG	10	,.,,.,,.,.	DABGIIIIII	5	,,.,.	>ABDA	6	C,,...	5G/BGB
7	.....,^].	DEGEGGE	10	,.,,.,,.,.	DABGIIIIII	5	,,.,.	>ABDA	6	.,,...	5GIBGB
7	.....,.	DEGEGGG	10	,.,,.,,.,.	DABGIIIIII	5	,,.,.	>ABDA	6	.,,...	5GIBGB
8	.....,.^],	DEGEGGGB	10	,.,,.,,.,.	DABGIIIIII	5	,,.,.	>ABDA	6	.,,...	5GIBGB
8	.....,.,	DEGEGGGB	10	,.,,.,,.,.	DABGIIIIII	5	,,.,.	>AB/A	6	.,,...	DGIBGB
8	.....,.,	DEGEGGGB	10	,.,,.,,.,.	DABGIIIIII	5	,,.,.	>ABAA	6	.,,...	DGIBGB
8	.....,.,	DEGEGGGB	10	,.,,.,,.,.	DABGIIIIII	5	,,.,.	>/BAA	6	.,,...	DGIBGB
9	.....,.,^],	DEGEGGGBE	10	,.,,.,,.,.	DABGIIIIII	5	,,.,.	>BBAA	6	.,,...	DGIBGB
9	.G...gG,,	DEGEGGGBG	10	,.,,.,,.,.	D3BGIIIIII	5	,,.,C	>BBAA	6	.,,...	DGIBGB
9	.....,.,,	DEGEGGGBG	10	,.,,.,,.,.	D3BGIIIIII	5	,,.,.	>BBAA	6	.,,...	/GIBGB

Wouldn't it be awesome if you knew what the frequency of C was in the population.

Bayesian inference

$$P(\theta|X) = \frac{P(X|\theta)P(\theta)}{P(X)} = \frac{P(X|\theta)P(\theta)}{\sum_{\theta} P(X|\theta)P(\theta)}$$



Distribution plots from Bink 2008

X = Data

θ = Parameter

Genotype posterior probabilities to improve genotyping accuracy

Using Bayes' Theorem, the posterior probability of genotype *g* is

$$\mathrm{P}\left(\mathrm{Genotype}=g\middle|\mathrm{Data}\right)=\frac{\mathrm{P}\left(\mathrm{Data}\middle|\mathrm{Genotype}=g\right)\mathrm{P}\left(\mathrm{Genotype}=g\right)}{\sum_{g\in\{0,1,2\}}\mathrm{P}\left(\mathrm{Data}\middle|\mathrm{Genotype}=g\right)\mathrm{P}\left(\mathrm{Genotype}=g\right)}$$

Genotype posterior probabilities to improve genotyping accuracy

Using Bayes' Theorem, the posterior probability of genotype *g* is

Likelihood of genotype *g* calculated as shown previously.

$$P(\text{Genotype}=g|\text{Data}) = \frac{P(\text{Data}|\text{Genotype}=g)\,P(\text{Genotype}=g)}{\sum_{g\in\{0,1,2\}} P(\text{Data}|\text{Genotype}=g)\,P(\text{Genotype}=g)}$$

Genotype posterior probabilities to improve genotyping accuracy

Using Bayes' Theorem, the posterior probability of genotype *g* is

Given an estimate of the population minor allele frequency, *f*, under HWE

$$P(\text{Genotype}=0|f)=(1-f)^2$$

$$P(\text{Genotype}=1|f)=2f(1-f)$$

$$P(\text{Genotype}=2|f)=f^2$$

Likelihood of genotype *g* calculated as shown previously.

$$P(\text{Genotype}=g|\text{Data}) = \frac{P(\text{Data}|\text{Genotype}=g)\,P(\text{Genotype}=g)}{\sum_{g\in\{0,1,2\}} P(\text{Data}|\text{Genotype}=g)\,P(\text{Genotype}=g)}$$

Genotype posterior probabilities to improve genotyping accuracy

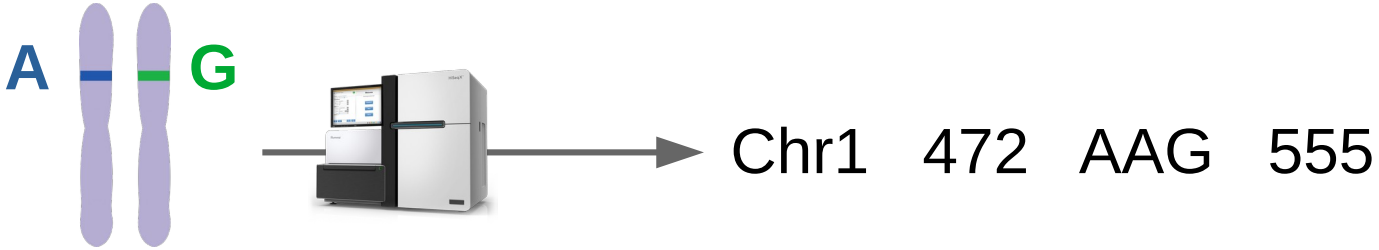
Using Bayes' Theorem, the posterior probability of genotype *g* is

Likelihood of genotype *g* calculated as shown previously.

Note: factors like inbreeding can easily be incorporated into the genotype posterior probabilities by conditioning on the allele frequency and inbreeding coefficient.

$$P(\text{Genotype}=g|\text{Data}) = \frac{P(\text{Data}|\text{Genotype}=g)\,P(\text{Genotype}=g)}{\sum_{g\in\{0,1,2\}} P(\text{Data}|\text{Genotype}=g)\,P(\text{Genotype}=g)}$$

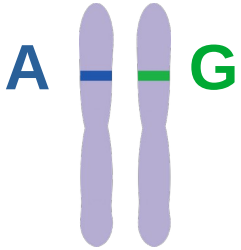
Example genotype posterior probability distribution



Assume we estimate f(A) = 0.7, f(G) = 0.3

Genotype	Log_{10} likelihood	Prior	Posterior probability
AA	-2.49	$P(\text{Genotype} = AA) = 0.7^2 = 0.49$	0.03
AG	-0.91	$P(\text{Genotype} = AG) = 2 \times 0.7 \times 0.3 = 0.42$	0.97
GG	-4.96	$P(\text{Genotype} = GG) = 0.3^2 = 0.09$	0.00

Example genotype posterior probability distribution





Chr1 472 AAG 555

Assume we estimate f(A) = 0.7, f(G) = 0.3

We could call the most probable genotype, AG, and have an associated degree of confidence (probability = 0.97).

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AG	-0.91	$P(\text{Genotype} = AG) = 2 \times 0.7 \times 0.3 = 0.42$	0.97
GG	-4.96	$P(\text{Genotype} = GG) = 0.3^2 = 0.09$	0.00

Exercise. Calculate genotype posterior probabilities and call genotypes with ANGSD.